Al Imam Mohammad Ibn Saud Islamic University
College of Computer and Information Sciences

Computer Science Department

Course Title:	Design and Analysis of Algorithms
Course Code:	CS 215
Course Instructor:	Prof Hachemi Bennaceur, Dr. Yassine Daadaa, Mrs Ghada AlOsaimi, Mrs Mai Al-Ammar
Exam:	Final exam
Semester:	Second Semester
Date:	8/05/2016 – 01/08/1437
Duration:	120 minutes
Marks:	40

Privileges: × Open Book × Open Notes ✓Calculator Permitted × Laptop Permitted

Student Name (in English):	**Key Answer**
Student ID:	
Section No.:	

Instructions:

1. Answer **5** questions; there are **5** questions in **8** pages.
2. Write your name on each page of the exam paper.
3. Write your answers directly on the question sheets. Use the ends of the question pages for rough work or if you need extra space for your answer.
4. If information appears to be missing from a question, make a reasonable assumption, state your assumption, and proceed.
5. **No** questions will be answered by the invigilator(s) during the exam period.
6. Provide **two** answers of one question will lose the grade of this question

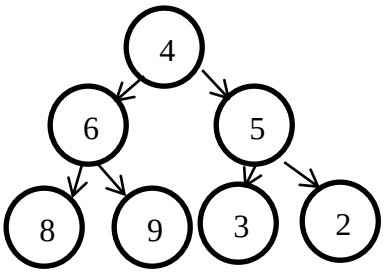
Official Use Only		
Question	**Student Marks**	**Question Marks**
1		10
2		8
3		8
4		6
5		8
Total		**40**

Student Name (in English): _____ Student ID: _____

Question 1: To be answered in (25) Minutes [ ] / 10 Marks

A. Choose the correct answer: (6 marks)

1. **Binary search algorithm is an example of algorithms that use:**
   a. Brute force design technique
   b. Divide and conquer design technique
   c. Dynamic programming design technique
   d. Iterative improvement design technique
2. **The node that violates a min-heap property is:**
   a. 4
   b. 5
   c. 6
   d. 9



3. **The main idea of the greedy algorithms is:**
   a. Searching for local optimal solutions
   b. Making local optimal solution in order to achieve global optimal solution
   c. Searching for all possible global optimal solutions
   d. Making local optimal solution in order to achieve a solution
4. **Finding the minimum number of wires that are used to connect a set of computers is an instance of :**
   a. Shortest path problem
   b. Minimum spanning tree (MST) problem
   c. 0/1 knapsack problem
   d. Fractional knapsack problem
5. **Which of the following algorithms is used to partition a graph into its strongly connected components?**
   a. Depth first search
   b. Breadth first search
   c. Prim's algorithm
   d. Dijkstra's algorithm

Student Name (in English): _____ Student ID: _____

6. **Which of the following is NOT correct about memoization:**
   a. It solves only sub-problems that are necessary
   b. It deals with overlapping sub-problems
   c. It improves performance because partial results are never calculated twice
   d. Every time it encounters a sub-problem even if it is new or seen before, it computes it again and stores the solution.

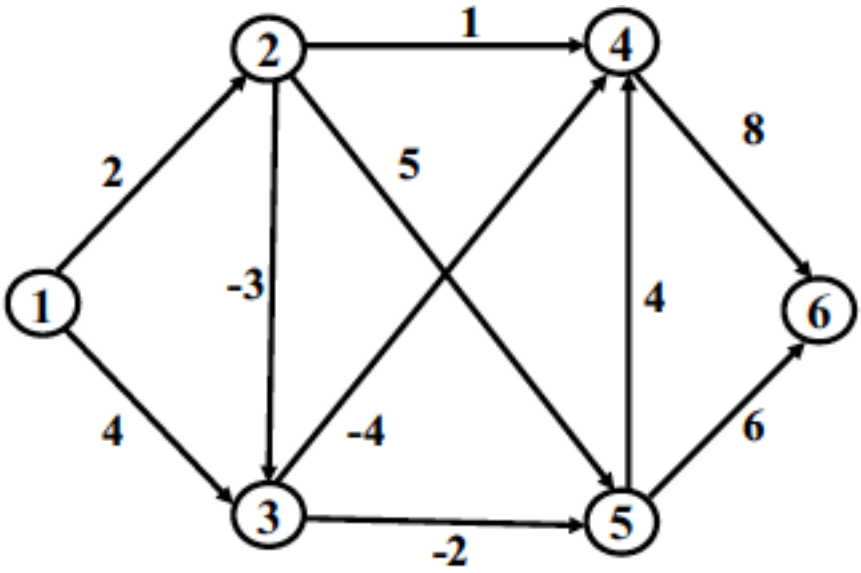
B. Complete the following table with the worst case running time for given algorithms: (4 marks)

Algorithm	**Worst case running time**
Insertion Sort	$O(N^2)$
Merge Sort	O(Nlogn)
Max Heapify	O(logn)
Bucket Sort	O(n)
Breadth First Search in a graph uses adjacency list	O(V+E)
Prim's algorithm for finding MST	O(ElogV)
Dijkstra algorithm to find shortest path	$O(\|V\|^2 + \|E\|)= O(\|V\|^2)$
0/1 Knapsack using Dynamic Programming	O(n * W)

Student Name (in English): _____ Student ID: _____

Question 2: To be answered in **(25) Minutes [ ] / 8 Marks**

Find the shortest paths from node 1 to every other node in the following graph using the Bellman-Ford algorithm. Show all the algorithm steps.



Solution:

	1	2	3	4	5	6
0	∞	∞	∞	∞	∞	∞
1	0	2	4	∞	∞	∞
2	0	2	-1	-5	-3	∞
3	0	2	-1	-5	-3	3
4	0	2	-1	-5	-3	3
5	0	2	-1	-5	-3	3

[initialization 0.5 and each value in the table except row 0 is 0.25]

Student Name (in English): _____ Student ID: _____

Question 3: To be answered in **(25) Minutes [ ] / 8 Marks**

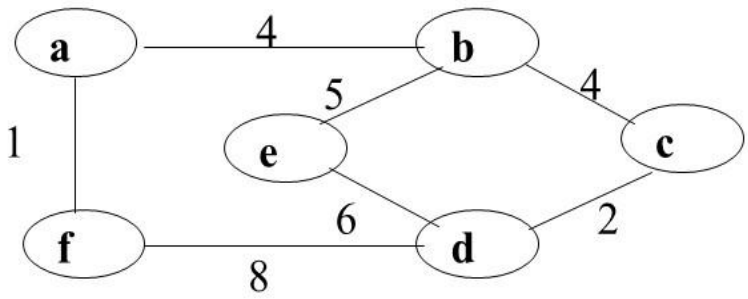
Consider the kruskal's algorithm seen in class:

```
Algorithm MCST-Kruksal (G, w)
{     T := Ø;
      for (each vertex v Є V[G]) do
          Make-Set(v);  // Make separate sets for vertices
      sort the edges by increasing weight w
      for (each edge (u,v) Є E, in sorted order) do
          if (Find-Set(u) ≠ Find-Set(v)) then
          {     // if no cycles are formed
                T := T U {(u,v)};       // Add edge to Tree
                Union(u,v);             // Combine Sets
          }
       return T;
}
```

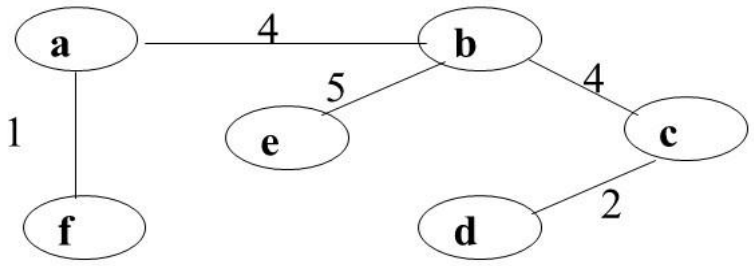
1. **Determine an efficient stopping criteria for this algorithm. (1 mark)**

   As any tree has n-1 edges (n is the number of vertices), then we can use a counter in the last loop ***for*** and when it reaches n-1 the algorithm can be stopped without considering all edges of the graph.

2. **Apply Kruskal's algorithm to find the minimum spanning tree of the following graph G, illustrate all steps, draw the minimum spanning tree and give its *cost*. (3.5 marks)**



Student Name (in English): _____ Student ID: _____

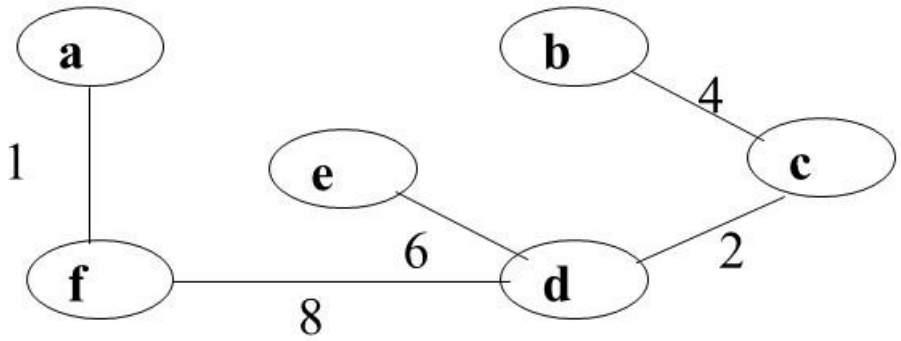


Cost = 16

[each edge 0.5 and cost 1]

3. **Using Kruskal's algorithm determine a spanning tree of G containing the edges (f,d), (e,d) and (b,c) with minimum cost. Show the first step, draw the spanning tree and give its *cost*. (3.5 marks)**

Step 1
Sets – {a}, {b, c}, {e}, {f, d, e}.
Take (c, d); Set(c) ≠ Set(d)
=> **Add** (c, d) to T,
Combine Set(c) & Set(d);
T = { f,d), (e,d), (b,c), (c,d) }.
Sets – {a}, {e}, {b, c, f, d, e}.



Cost =21.

Student Name (in English): _____ Student ID: _____

Question 4: To be answered in (20) Minutes [ ] / 6 Marks

Consider the following instance of the 0,1 knapsack problem:

Item	**1**	**2**	**3**	**4**
Benefit	6	12	4	9
Weight	1	5	3	4

The maximum allowable total weight in the knapsack is W = 5.
Find an optimal solution for the above problem with the weights and benefits above using Dynamic Programming. Be sure to state both the *value* of the maximum benefit that you obtain as well as the *item(s)* that you need to obtain this benefit. Show all steps.

Solution [each row in value table except first one is 1 mark, max value 0.5, each item of selected items 0.75]
Value Table:

Value	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	6	6	6	6	6
2	0	6	6	6	6	12
3	0	6	6	6	10	12
4	0	6	6	6	10	15

Maximum Benefit = Value[4][5] = 15
To determine which items in knapsack :
Start with i=4 and w = 5
V[4,5] ≠ V[3,5] ➔ include item 4 in knapsack , i=3, w=5-4 = 1
V[3,1] = V[2,1] ➔ ignore item 3 , i=2
V[2,1] = V[1,1] ➔ ignore item 2 , i=1
V[1,1] ≠ V[0,1] ➔ include item 1 in knapsack , i=0, w = 1-1 = 0 , stop here

Items in knapsack is items 1 and 4

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Question 5: To be answered in **(25) Minutes [ ] / 8 Marks**

Assume that you are given a chain of matrices < A1, A2, A3, A4 >, with dimensions 2×5, 5 × 4, 4 × 2 and 2 × 4 respectively, M and S tables. Answer the questions below:

M [i,j]

	A1	A2	A3	A4
A1	0	40	56	72
A2		0	40	?
A3			0	?
A4				0

S [i,j]

	A1	A2	A3	A4
A1	0	1	2	3
A2		0	2	?
A3			0	3
A4				0

1. Which class of algorithm (or technique) can be used to compute the optimal number of multiplications required to calculate the chain product? (1 mark)
   Dynamic Programing

2. How many different ways are there to parenthesize the chain of four matrices? (1 mark)
   There are five ways to fully parenthesize chain product

3. Matrix-chain multiplication algorithm uses two tables M and S to fully parenthesize the chain product. What is the kind of information provided by these tables? (2 mark)
   M is used to find minimum number of scalar multiplications of matrix-chain (or find optimal cost of parenthesizing the chain product) and S is used to find value of K at which splits the product in optimal parenthesization

4. What is the optimal number of multiplications required to calculate the chain product <A2, A3, A4 >? (1 mark)
   80

5. What is the optimal number of multiplications required to calculate the chain product < A3, A4 >? (1 mark)
   32

6. What is the value of K that splits the chain product < A2, A3, A4 > in an optimal parenthesization? (1 mark)
   3

7. Find the optimal parenthesization of multiplying four matrices. (1 mark)
   (((A1 A2) A3) A4)

Good Luck